

2025 WINTER GLOBAL PROGRAM

SOUND BUDDY

Create Value through Noise Visualization
Smart City Solution with Machine Learning

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Collaboration with Bellevue College (US)

EXISTING FOCUS

Public Safety

🛡️ Gunshots & Emergencies

⚠️ Critical but Rare

For Police / Gov



NEW TARGET

Asset Value

🏠 Property Valuation

📈 Frequent & Daily Impact

👥 For Everyone (Huge Market)

The Invisible Value: NOISE

When buying a house, we check everything.
But why do we still rely on "**Gut Feelings**" for the most critical factor?

HOME BUYING CHECKLIST

 Traffic & Accessibility

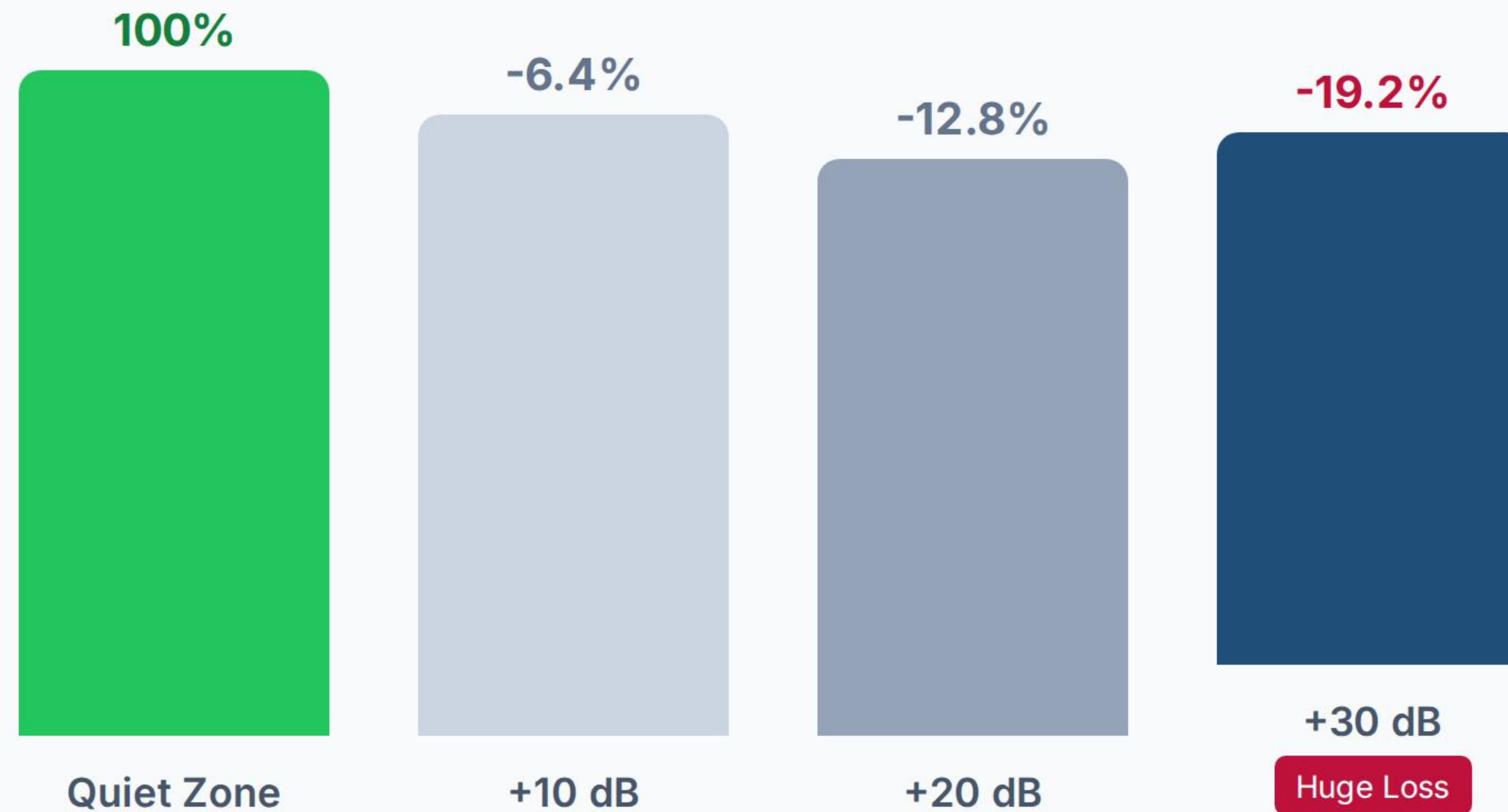
 School District

 Sunlight & View

 Quietness & Noise ?

**"For every 1dB increase in noise,
house prices decrease by 0.64%."**

Source: Theebe (2004), *The Journal of Real Estate Finance and Economics*



1. PERSONA ANALYSIS

Three Key Targets derived from Data



Emily (28)

Remote Software Engineer

"Noise isn't just annoying; it costs me my salary."

NEEDS & PAIN POINTS

- Absolute quiet during work (9-6)
- 23 mins to regain focus after distraction
- Needs noise data before moving



Sophia (34)

Stay-at-home Mom

"I need to know instantly if that loud bang is a truck or a gunshot."

NEEDS & PAIN POINTS

- Instant alerts for dangerous sounds
- Protect child's nap time (Predictability)
- WHO: Noise harms cognitive development



Robert (67)

Retired Teacher (Hypertension)

"Is the neighborhood getting louder, or am I just getting old?"

NEEDS & PAIN POINTS

- Long-term noise risks heart health
- Needs objective long-term noise trends
- Finding a quiet place for retirement

2. HYPOTHESIS

Dual Approach Strategy: Macro (Moving) & Micro (Living)



MACRO SCENARIO (MOVING)

Uncertainty Reduction Hypothesis

"Before signing a lease, people will **pay a premium** to eliminate the anxiety of unknown noise risks."



Target: CTR +20% for Listings with Reports



MICRO SCENARIO (LIVING)

Active Avoidance & Discovery Hypothesis

"If residents receive noise forecasts, they will take **active measures**, such as moving to quieter spots like cafes."



Target: 30% Behavior Change Rate

3. MVP MODEL

Sound Buddy Lite: Maximizing Value with Minimal Features

Hybrid MVP Strategy

Prototype delivering core value before full app development

Premium Insight Report

Target: Moving (Macro)

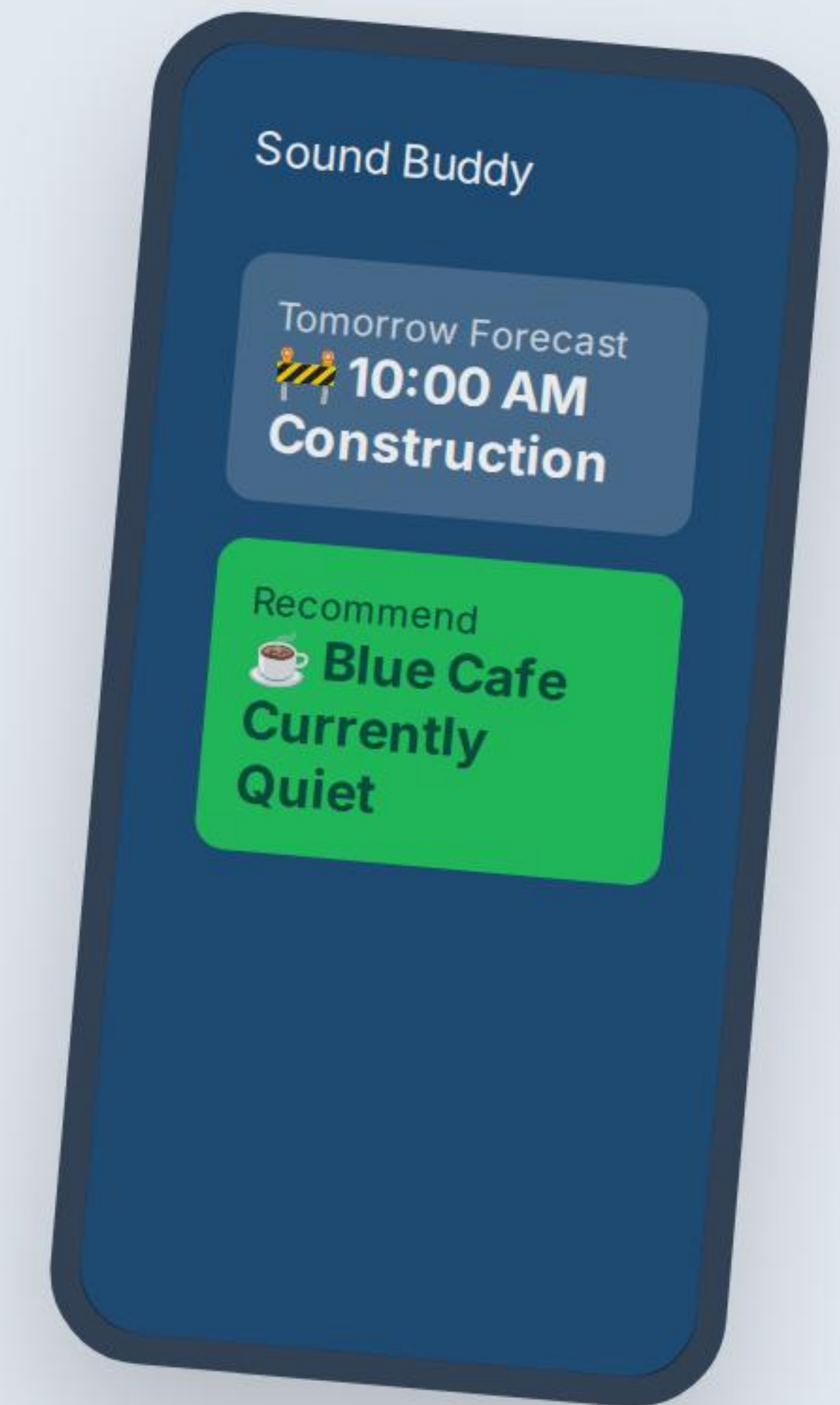
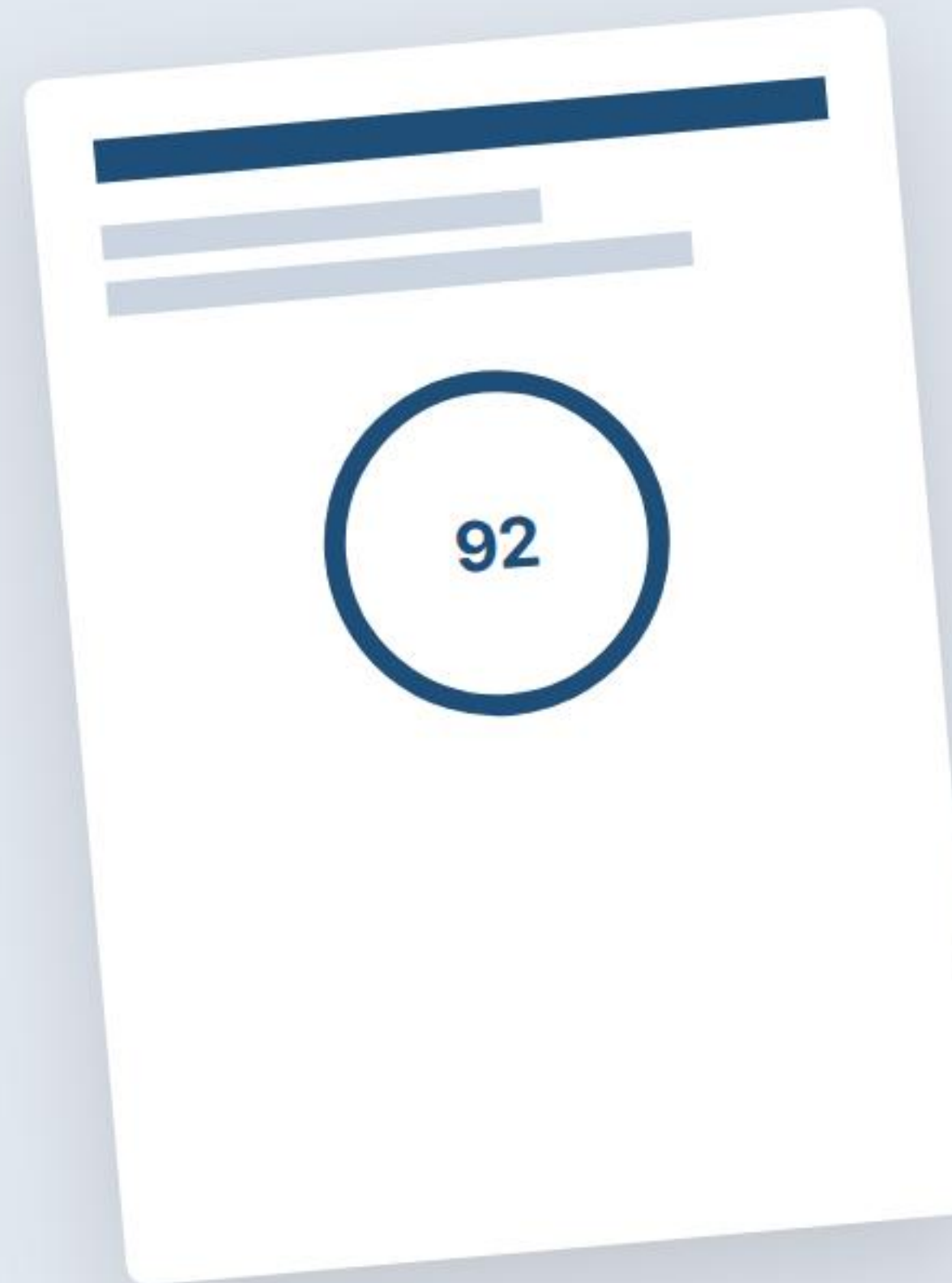
A single-page PDF report inserted into real estate listings.
Visualizes '9-to-6 Quiet Certification' & 'Safety Score'.

Daily Noise Alert

Target: Living (Micro)

Alerts like "Construction at 10 AM tomorrow" and
recommendations for currently quiet cafes.

Paper Prototype & App Service



4. VALIDATION PLAN

Experimental Design to Prove Hypotheses

PHASE 1
MARKET DEMAND



Real Estate Value

Method: Smoke Test (A/B Test)

Post listings with and without the 'Noise Insight Badge' on real estate sites to compare engagement.

**Target:
CTR +20%**

PHASE 2
DAILY USAGE



Daily Utility

Method: Wizard of Oz Test

Group chat with 50 students. Manually send noise forecasts and quiet spot recommendations to track movement.

**Target:
Behavior Change 30%**

PHASE 3
TECH FIT



Willingness to Pay

Method: Scenario Survey

Interview personas to determine willingness to pay extra rent for listings with verified noise reports.

**Target:
60% answer 'Yes'**